

Virtual classroom for ET4icp simulations

The virtual classroom will be a Teams environment, for which we will provide a link. All the students of the group will call in and so will one of the TA's for online support. Next, one of the students (I would recommend the ones with the fastest internet connection) will login to the simulations server and share her/his desktop so that all the students can see the output of the simulations. This student is in control of running the simulations, although you can take over control in Teams using the 'Request control' option.

This manual provides a short guide illustrating how to log in to the server, how to navigate, how to edit files, and finally how to save graphs. The manual only provides support for Windows, using a Mac is at your own risk and no support can be given due to the unavailability of hardware. **It is recommended that all students follow the preparations so that another student can log in to the server in case there are problems.**

Preparations

1. Install a SSH client with X11 support, the recommended and tested client is MobaXTerm (free version): <https://mobaxterm.mobatek.net/download.html>
2. Install the TUDelft VPN client and set it up according to the manual: <https://www.tudelft.nl/en/it-manuals/applications/vpn>

Logging in

1. First start the VPN connection to TUDelft
2. Start MobaXTerm and open a SSH session to **et4icp.ewi.tudelft.nl**, see also the screenshot below (click the button Session)

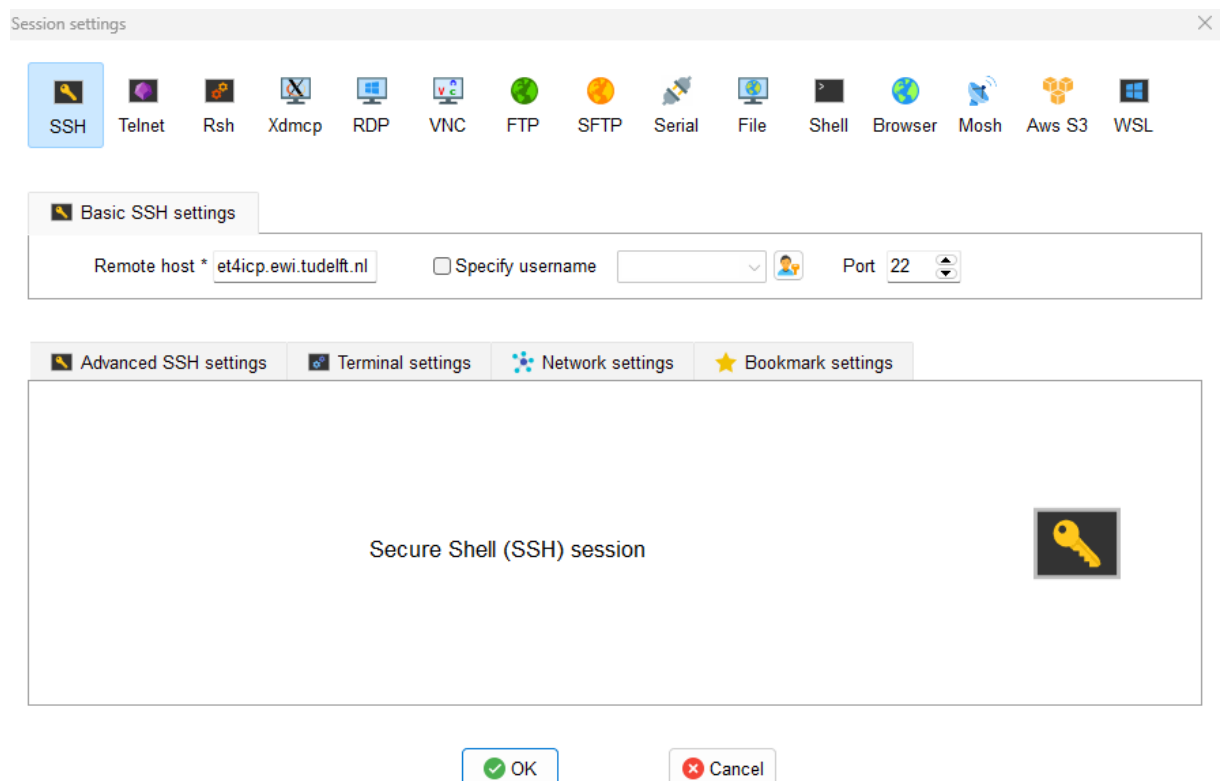
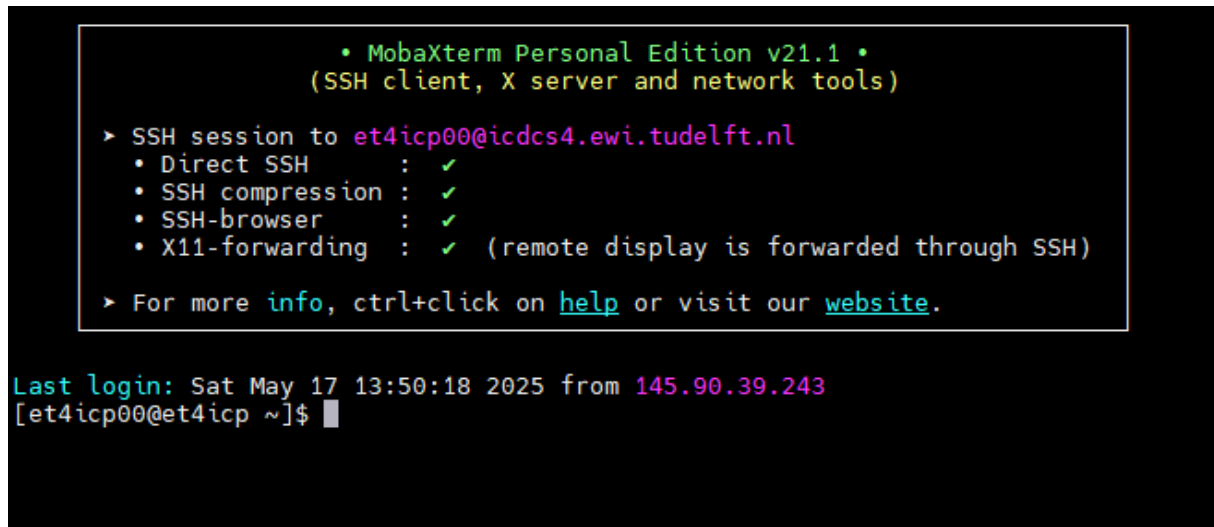


Figure 1: MobaXTerm SSH session

3. Once you start the connection click OK if you get a security warning asking you to verify the server ID.
4. Login using the account name and password will provided by the TA.
5. You should now see the command prompt as shown in the figure below.



```

• MobaXterm Personal Edition v21.1 •
(SSH client, X server and network tools)

> SSH session to et4icp00@icdcs4.ewi.tudelft.nl
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✓ (remote display is forwarded through SSH)

> For more info, ctrl+click on help or visit our website.

Last login: Sat May 17 13:50:18 2025 from 145.90.39.243
[et4icp00@et4icp ~]$

```

Figure 2: Prompt of command window after login

Navigating and editing

- The command prompt is Linux Bash
- Navigating to a directory is done with the command **cd your/folder/bla**, to go back you can use **cd ..** to go back one level, or **cd ~** to go back to your home directory (/home/ictprak1)
- You can use the tab key on your keyboard for autocomplete when typing a command
- Pressing the arrow up key displays the previous command
- The content of a directory can be shown with the command **ls**
- You can edit with the Nano text editor using **nano filename.ext**, you should get the screen below (fig. 3) in case you opened implant.sub
- To save in Nano press Ctrl+o, to exit press Ctrl+x
- You can also navigate in MobaXTerm by processing the >> button at the left side of the screen and selecting the Scp tab. Please note that this navigates separately from the command prompt

Running simulations

- First set the simulation environment and copy the base files by running the command: **source /users/et4icp/ET4ICP_2025/initialSetup.sh**
- If the source directories already exist, it will ask if you want to overwrite the files. Do not do this if you already ran parts of the simulations
- The commands are the same as in the manual, e.g. **sprocess -n** for SProcess and **swb &** for the Sentaurus Workbench. The & launches the workbench while allowing you to still use the terminal
- At the end of the sprocess simulation a window should pop up with a graph of the simulation results. You will also see a warning message, but this can be ignored
- The sprocess scripts also create a CSV .out file for plotting using your favourite software package for making graphs (hint: avoid Excel). In Sentaurus Workbench .png files of important results are automatically generated when SVisual is ran.

- After closing a GUI window, you will again see an error message in the command prompt that you can ignore. Press enter to return to the prompt.

Wrapping up

1. Make sure you have saved all results on your own PC. We will change the passwords daily, hence after the session you will not be able to login again
2. Logout by typing **logout**
3. Finally you can close the VPN connection after you have closed the Teams session (otherwise you will lose the Teams connection).

```

GNU nano 5.6.1                                implant-As.cmd
##### GRID DEFINITION #####
line x loc=0<um> spacing=1<nm> tag=left
line x loc=1.0<um> spacing=1<nm> tag=right
##### STARTING MATERIAL #####
region Silicon x1=left x2=right
init concentration=1e14<cm-3> field=Boron
AdvancedCalibration
##### IMPLANTATION #####
implant Arsenic dose=1e15 energy=148
##implant Antimony dose=1e15 energy=288
##implant Boron dose=1e15 energy=24
## implant Phosphorus dose=1e15 energy??
##### PLOTTING #####
select z=log10(Arsenic)
plot.id color=black title="Implantation" label="Arsenic" min = {0 14} max = {1 20}
##select z=log10(Antimony)
##plot.id !clear color=red label="Antimony"
##select z=log10(Boron)
##plot.id !clear color=green label="Boron"
##select z=log10(Phosphorus)
##plot.id !clear color=blue label="Phosphorus"
#####
## Export data to CSVs
print.data outfile=implantAs.out name=Arsenic
##print.data outfile=implantB.out name=Antimony
##print.data outfile=implantB.out name=Boron
##print.data outfile=implantP.out name=Phosphorus

^G Help      ^O Write Out  ^W Where Is   ^X Cut        ^_ Execute   ^L Location  ^-U Undo     ^M Set Mark   ^-] To Bracket ^-O Previous  ^B Back      ^_ Prev Word
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify   ^-G Go To Line ^-R Redo     ^M-C Copy     ^-; Where Was ^-N Next     ^F Forward   ^_ Next Word
  
```

Figure 3: Screenshot of the Nano text editor